



Translated

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# Josephus Problem

## Statement

We are given the natural numbers  $n$  and  $k$ . All natural numbers from  $1$  to  $n$  are written in a circle. First, count the  $k$ -th number starting from the first one and delete it. Then  $k$  numbers are counted starting from the next one and the  $k$ -th one is removed again, and so on. The process stops when one number remains. It is required to find the last number.

This task was set by **Flavius Josephus** in the 1st century (though in a somewhat narrower formulation: for  $k = 2$ ).

This problem can be solved by modeling the procedure. Brute force modeling will work  $O(n^2)$ . Using a [Segment Tree](#), we can improve it to  $O(n \log n)$ . We want something better though.

## Modeling a $O(n)$ solution

We will try to find a pattern expressing the answer for the problem  $J_{\{n, k\}}$  through the solution of the previous problems.

Using brute force modeling we can construct a table of values, for example, the following:

```

\begin{array}{cccccccccc}
n \setminus k & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
2 & 2 & 2 & 1 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\
3 & 3 & 3 & 3 & 2 & 2 & 1 & 3 & 3 & 2 & 2 \\
4 & 4 & 4 & 1 & 1 & 2 & 2 & 3 & 2 & 3 & 3 \\
5 & 5 & 5 & 3 & 4 & 1 & 2 & 4 & 4 & 1 & 2 \\
6 & 6 & 6 & 5 & 1 & 5 & 1 & 4 & 5 & 3 & 5 \\
7 & 7 & 7 & 7 & 4 & 2 & 6 & 3 & 5 & 4 & 7 \\
8 & 8 & 8 & 1 & 7 & 6 & 3 & 1 & 4 & 4 & 8 \\
9 & 9 & 9 & 3 & 1 & 1 & 8 & 7 & 2 & 3 & 8 \\
10 & 10 & 5 & 4 & 5 & 3 & 3 & 9 & 1 & 7 & 8
\end{array}

```

And here we can clearly see the following **pattern**:

$$J_{\{n, k\}} = \left( (J_{\{n-1, k\}} + k - 1) \bmod n \right) + 1$$
$$J_{\{1, k\}} = 1$$

Here, 1-indexing makes for a somewhat messy formula; if you instead number the positions from 0, you get a very elegant formula: